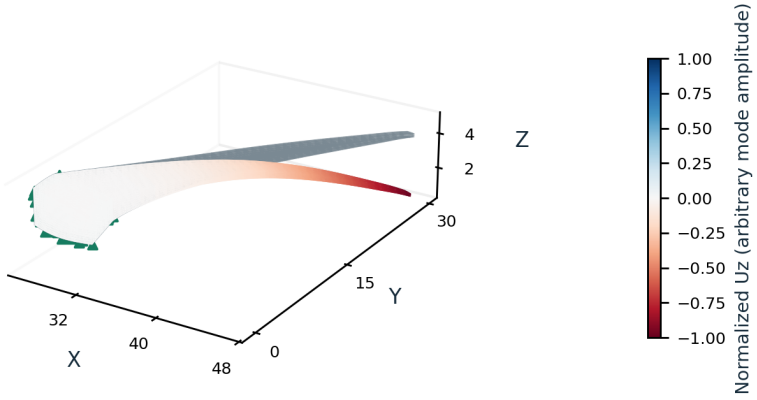
CRM wingbox: first five modes

CRM_wingbox_modal | SOL 103 | Reproducibility / historical regression

Public TACS CRM geometry; isotropic conversion E=70 GPa, nu=0.3, rho=2500 kg/m3, t=0.02 m.

Geometry / mesh	23738 nodes; 25055 elements. Public TACS CRM geometry; isotropic conversion E=70 GPa, nu=0.3, rho=2500 kg/m3, t=0.02 m.
Material / section	E=7e+10, nu=0.3; density=2500; shell thickness=0.02. SI: m, N, kg, s, Pa
Boundary conditions	Upstream root and side-of-body constraints retained; lumped shell mass (cardless default); PARAM K6ROT 100. Adapted CQUAD4 model.
Applied load	Free vibration about the undeformed state; no applied static preload.

Mode 1 | f = 1.6827 Hz



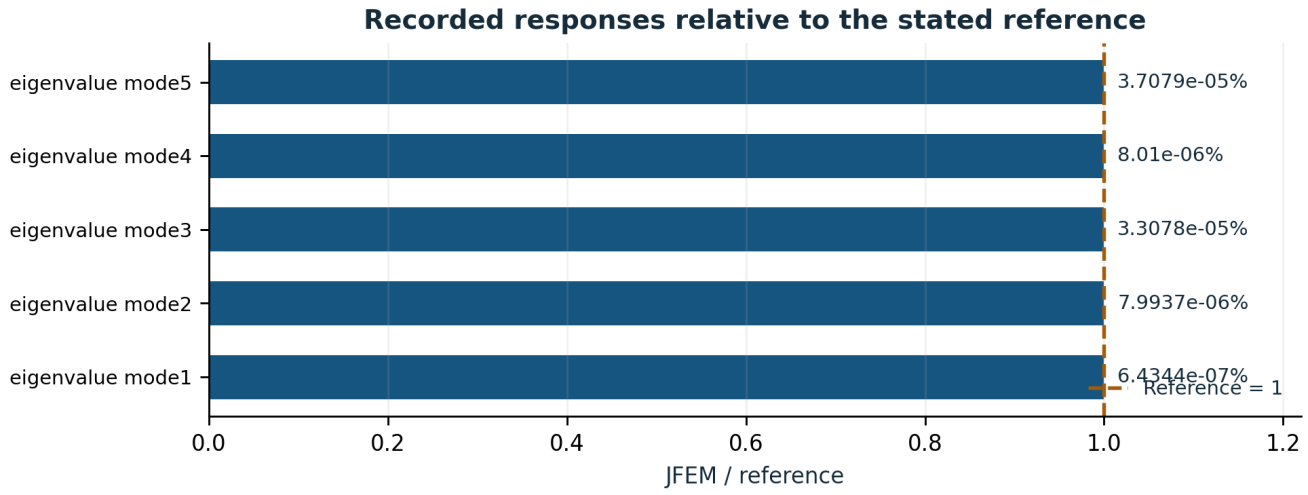
Shape amplified x111; grey = original. Eigenvector sign and amplitude are arbitrary. All faces shown; deformed edges hidden.

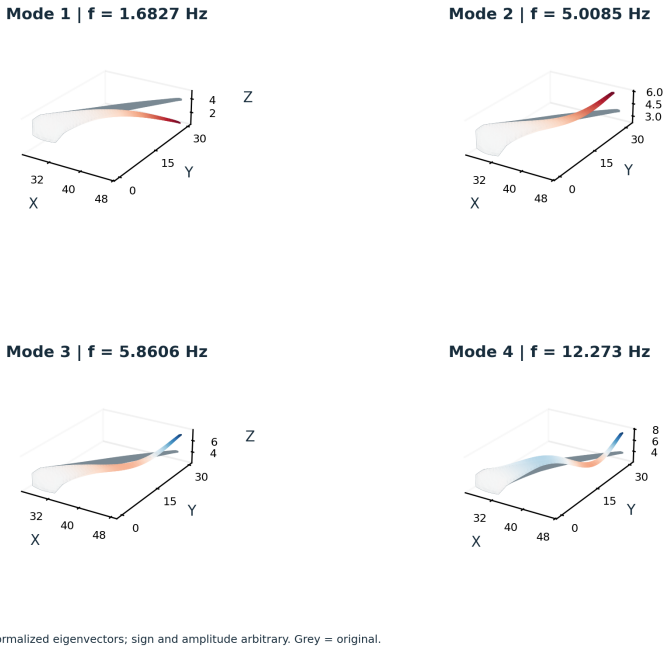
Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
eigenvalue mode1 [rad^2/s^2]	111.78	111.78	6.4344e-07%	REGRESSION
eigenvalue mode2 [rad^2/s^2]	990.31	990.31	7.9937e-06%	REGRESSION
eigenvalue mode3 [rad^2/s^2]	1355.9	1355.9	3.3078e-05%	REGRESSION
eigenvalue mode4 [rad^2/s^2]	5946.7	5946.7	8.01e-06%	REGRESSION
eigenvalue mode5 [rad^2/s^2]	10449	10449	3.7079e-05%	REGRESSION

Evidence, convergence and interpretation





Independently normalized eigenvectors; sign and amplitude arbitrary. Grey = original.

First available modes. Red/blue indicate opposite signed components; each mode is normalized independently.

Stored historical scalar has no independently reproducible public source. This comparison is a regression check and is not independent scientific validation.

Scope: Public geometry with synthesized isotropic properties. The listed historical eigenvalues have no independently published matching target; this case demonstrates industrial model scale and reproducibility.

Reference and reproduction

TACS public CRM example (Apache-2.0): source geometry and material/thickness definitions. The stored frequencies are not independently published validation targets. https://github.com/smdogroup/tacs/blob/master/examples/crm/crm_frequency.py

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv